

# GNU-Plot-Lab-Notebook

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## Question 1

Plot the following functions for  $-10 \leq x \leq 10$  on the same graph using GNUPLOT. Set 1000 samples. Take the range of  $y$ -values from  $-4$  to  $+4$ . Mark each curve with different line styles and line widths. Label  $x$ -axis and  $y$ -axis as ' $x$ -values' and ' $y$ -values' respectively. Set the title of the plot as 'Function plots'. Save the plot in a suitable format.

$$f_1(x) = \coth(x) - \frac{1}{x}, \quad f_2(x) = e^{-\sin(x)}, \quad f_3(x) = x|x|$$

**Sol:** The GNU-Plot code for the solution of this problem is given below.

```
1  reset
2  set term png
3  set ou "C:/Users/Suparnajit2005/Documents/GitHub/SEC-lab/GNU-plot/Lab-files/Output
   /1/Q_xxi_ou.png"
4
5  set grid
6  set xzeroaxis linestyle 1 lc "black"
7  set yzeroaxis linestyle 1 lc "black"
8  set xrange [-10:10]
9  set yrange [-4:4]
10 set xlabel "x values"
11 set ylabel "y values"
12 set samples 1000
13
14 f1(x)=(1/(tanh(x)))-(1/x)
15 f2(x)=exp(-sin(x))
16 f3(x)=x*abs(x)
17
18 plot f1(x) lw 2 lc "#5f0f40" dt 1 , f2(x) lw 2 lc "#fb8b24" dt 2, f3(x) lw 2 lc "
   #9a031e" dt 3
```

The output plot is as follows,